# Multi-Disease Detection Report

CNN + LLM contextual reasoning • generated 2026-08-24 22:58

## Patient context

| | |
|---|---|
| **Age** | 22 |
| **Sex** | Male |
| **Symptoms** | Persistent fever, fatigue, easy bruising, frequent infections, bone pain, swollen lymph nodes |
| **History** | 22-year-old patient, symptoms worsening rapidly over 6 weeks, recurrent infections and unexplained bruising |

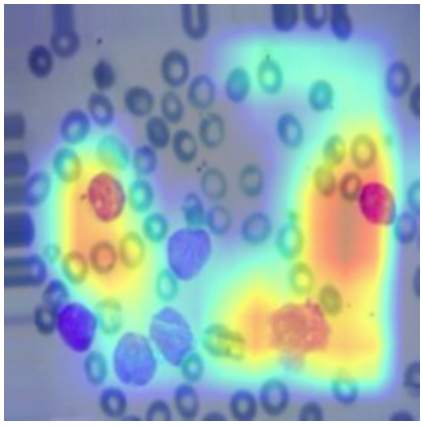
## Model prediction

**all_pro** **Malignant**

Acute Lymphoblastic Leukemia (ALL) · Hematologic / Bone marrow & blood · Peripheral blood smear microscopy

| # | Condition | Confidence |
|---|---|---|
| 1 | all_pro | 93.6% |
| 2 | kidney_tumor | 1.3% |
| 3 | kidney_normal | 1.0% |
| 4 | cervix_dyskeratotic | 1.0% |
| 5 | all_pre | 0.8% |

## Grad-CAM (model attention)



## LLM contextual interpretation

**Model suggests Acute Lymphoblastic Leukemia (ALL) (all_pro) at 93.6% confidence.**

**Model finding.** The CNN's top class is 'all_pro' (Malignant, Hematologic / Bone marrow & blood) on Peripheral blood smear microscopy, at 93.6%.

**Confidence.** High model confidence (93.6%) with a clear 92.4 pt margin over the runner-up.

**Differential**

- **all_pro** — ranked #1 at 93.6%
- **kidney_tumor** — ranked #2 at 1.3%
- **kidney_normal** — ranked #3 at 1.0%

**Grad-CAM correlation.** Model attention was a moderately sized region in the central-right portion of the image (~22% high-salience area). Verify this region is anatomically plausible for Peripheral blood smear microscopy.

**Biomedical context.** [KB] Pro-B stage lymphoblasts consistent with more advanced ALL. Group: Acute Lymphoblastic Leukemia (ALL); organ: Hematologic / Bone marrow & blood; modality: Peripheral blood smear microscopy; category: Malignant (severity: high). ALL is characterized by lymphoblast proliferation in marrow and blood; morphologic stages and immunophenotyping (pre-B/pro-B) guide risk stratification. [BioGPT fallback — set ENABLE_BIOGPT=true to run the real model] Diagnosis is confirmed by marrow aspirate/biopsy with flow cytometry and cytogenetics; blast percentage and immunophenotype guide risk stratification. [Med-PaLM 2: PLACEHOLDER — no public API; see medpalm_adapter.py to wire MedLM]

**Patient considerations.** Provided context (age: 22; sex: Male; symptoms: Persistent fever, fatigue, easy bruising, frequent infections, bone pain, swollen lymph nodes; history: 22-year-old patient, symptoms worsening rapidly over 6 weeks, recurrent infections and unexplained bruising) should be correlated with the finding.

**Recommended next steps**

- Correlate with the full clinical picture and prior imaging.
- Obtain gold-standard confirmation appropriate to the modality (e.g., biopsy/histology or specialist review).
- Escalate to the relevant specialist if the category is malignant or high-severity.

**Uncertainty & limitations**

- Interpretation generated WITHOUT the Claude reasoning model (offline fallback).
- Model trained on specific public datasets; performance may not transfer to other scanners/populations.
- A confident softmax score is not calibrated probability of disease.

■ Research decision-support only — not a medical diagnosis. All findings require confirmation by a licensed clinician.